

CFD Study & Validation of Turbulent Pipe Flow

Turbulence Model and Y+ Study
Validation of Various Empirical Correlations

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Problem Description

- Flow Regime & Geometry:
- Turbulent flow simulation at $Re = 10,000$ in a pipe of length $L = 16$ m and diameter $D = 0.2$ m ($L/D = 80$).
- Fluid Properties: Operating velocity $V = 1.0$ m/s, fluid density $\rho = 1,000$ kg/m³, and dynamic viscosity $\mu = 0.02$ Pa·s.
- **Key Objectives:**
 - Compute and analyze radial velocity profiles, axial pressure drop, Darcy friction factor (f), and skin friction coefficient (C_f).
 - Validate numerical results against analytical solutions and empirical correlations.
 - Post-process the flow field using velocity vectors, pressure contours, and wall Y^+ distributions.

$$Re = \frac{\rho V D}{\mu}$$

$$Cf = \frac{f}{4}$$

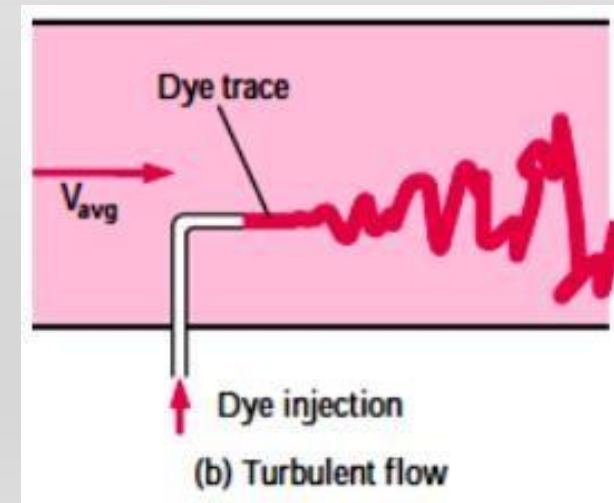
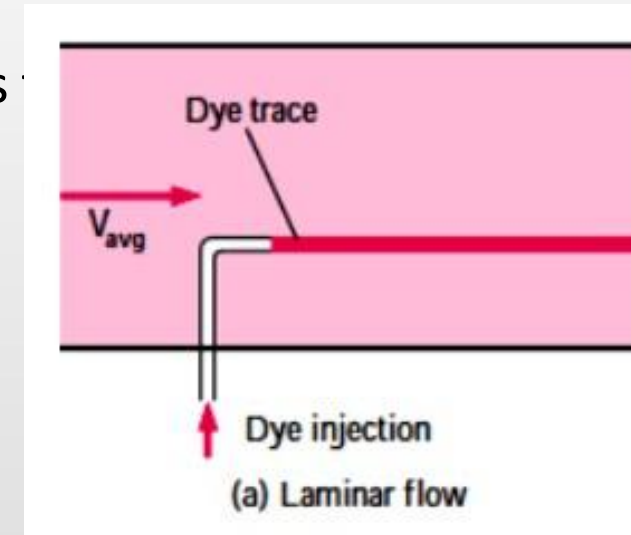
$$C_f = \frac{\tau_w}{\frac{1}{2} \rho V_{avg}^2}$$

Turbulent Pipe Flow - Theory

- Flow can be laminar or turbulent depending on Reynolds number, which is forces to the viscous forces.

$$Re = \frac{\rho v^2 L^2}{\mu v L} = \frac{\rho v L}{\mu}$$

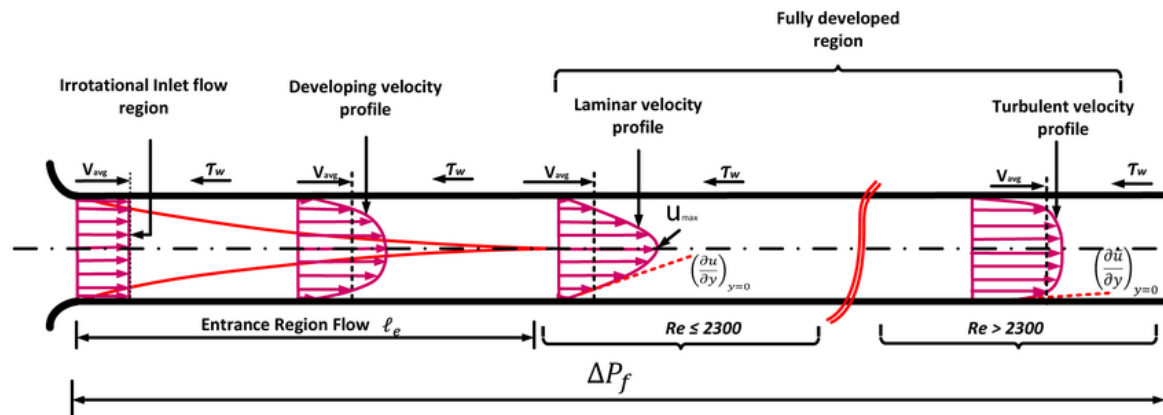
- Flow is laminar for $Re < 2300$, turbulent for $Re > 4000$
- Laminar flow is stream line flow and every layer of flow is moving in well defined manner. However in turbulent flow, it is characterized by chaotic movement.



Turbulent Pipe Flow - Theory

Fully developed flow and entrance length

- Flow is said to be fully developed when axial velocity does not change with axial location.



$$\frac{\partial u(r, x)}{\partial x} = 0 \rightarrow u = u(r)$$

$$L_{h, \text{turbulent}} = 1.359 D Re_D^{1/4}$$

These correlations are just approximations, practically values may vary with large margin. But these correlations provides you the right direction and guide you what results should be look like.

$$L = 16 \text{ m}, D = 0.2 \text{ m}$$

For practical CFD, entrance length will vary.

For current case, flow will be fully developed after $L/D = 13.59$ or $L = 13.59 * 0.2 = 2.718 \text{ m}$

Analytical and semi-empirical approach

- For current case with $Re = 10000$, density = 1000 Kg/m^3 , dia $D = 0.2 \text{ m}$, $V_{\text{mean}} = 1 \text{ m/s}$ and dynamic viscosity $\mu = 0.02 \text{ Pa.S}$ and pipe length = 16 m .

$$L_{h, \text{turbulent}} = 1.359 D Re_D^{1/4}$$

- Flow will be fully developed (theoretically) at $L/D = 13.59$ or $L = 13.59 * 0.2 = 2.718 \text{ m}$

- Friction factor for smooth pipe for current case $f = 0.030886$

$$Cf = \frac{0.030886}{4} = 0.0077215$$

$$\frac{1}{\sqrt{f}} \cong -1.8 \log \left[\frac{6.9}{Re} + \left(\frac{\epsilon/D}{3.7} \right)^{1.11} \right]$$

(you can also find value of friction factor from moody diagram for given Reynolds number and roughness...)

$$Cf = \frac{\tau_w}{\frac{1}{2} \rho V_{avg}^2} \quad Cf = \frac{f}{4}$$

- I can find out pressure drop/loss from Darcy formula

$$\text{Pressure drop} = 0.030866 \times \frac{1000 \times (1)^2}{2} \times \frac{16}{0.2} = 1234.64 \text{ Pa}$$

$$\Delta P = f \frac{\rho u^2 l}{2d}$$

Analytical and semi-empirical approach

$$P_1 - P_2 = \rho g h_L$$

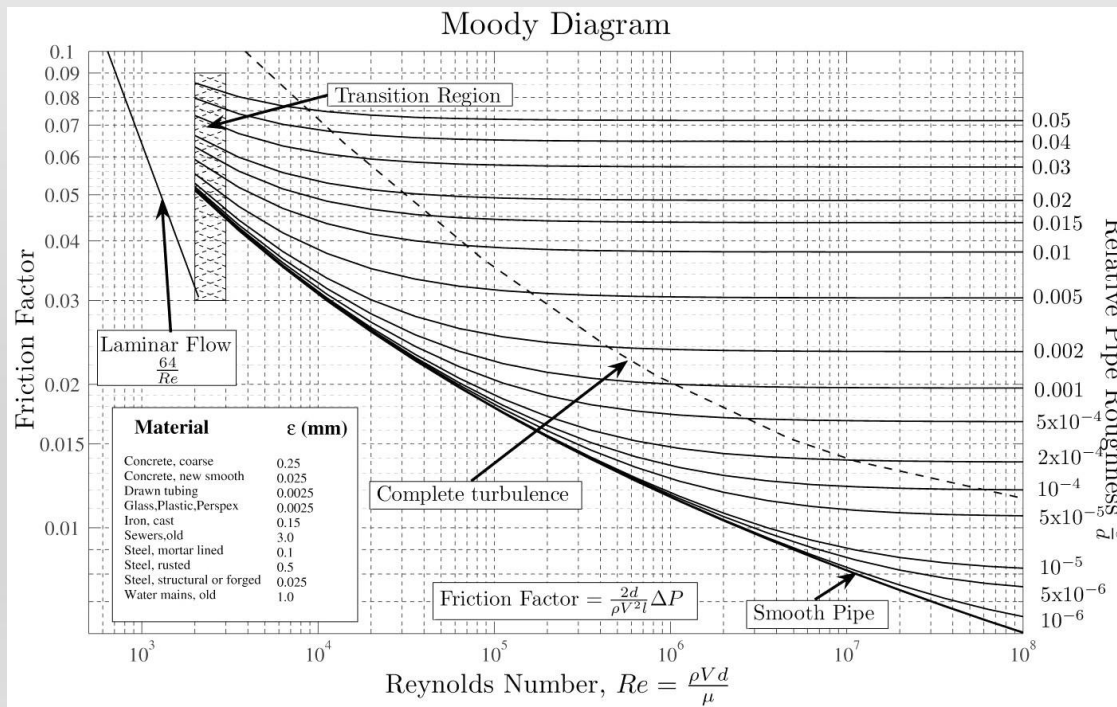
$$P_1 - P_2 = 4\tau_w \frac{L}{D}$$

$$h_L = \frac{4\tau_w}{\rho g} \frac{L}{D}$$

$$h_L = f \frac{L}{D} \frac{V^2}{2g}$$

$$f = \frac{8\tau_w}{\rho V^2}$$

$$\epsilon/D = \text{roughness factor}$$



Colebrook equation

$$\frac{1}{\sqrt{f}} = -2.0 \log \left(\frac{\epsilon/D}{3.7} + \frac{2.51}{Re \sqrt{f}} \right)$$

S.E. Haaland 1983

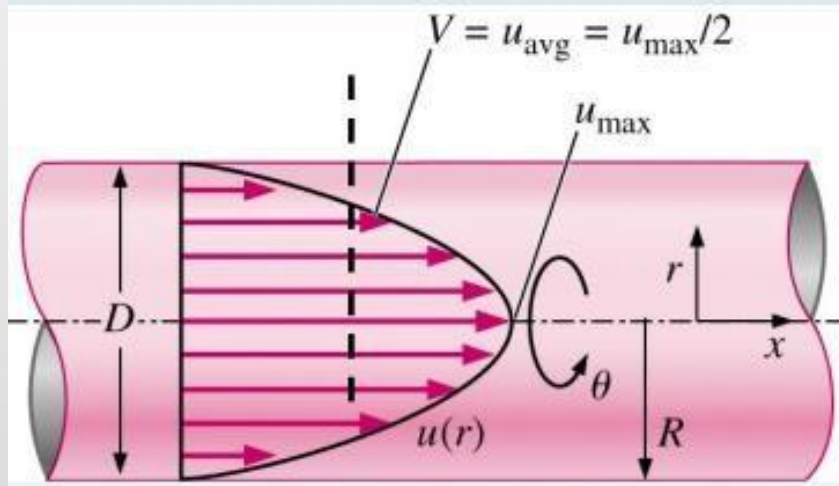
$$\frac{1}{\sqrt{f}} \cong -1.8 \log \left[\frac{6.9}{Re} + \left(\frac{\epsilon/D}{3.7} \right)^{1.11} \right]$$

Relative Roughness, ϵ/D	Friction Factor, f
0.0*	0.0119
0.00001	0.0119
0.0001	0.0134
0.0005	0.0172
0.001	0.0199
0.005	0.0305
0.01	0.0380
0.05	0.0716

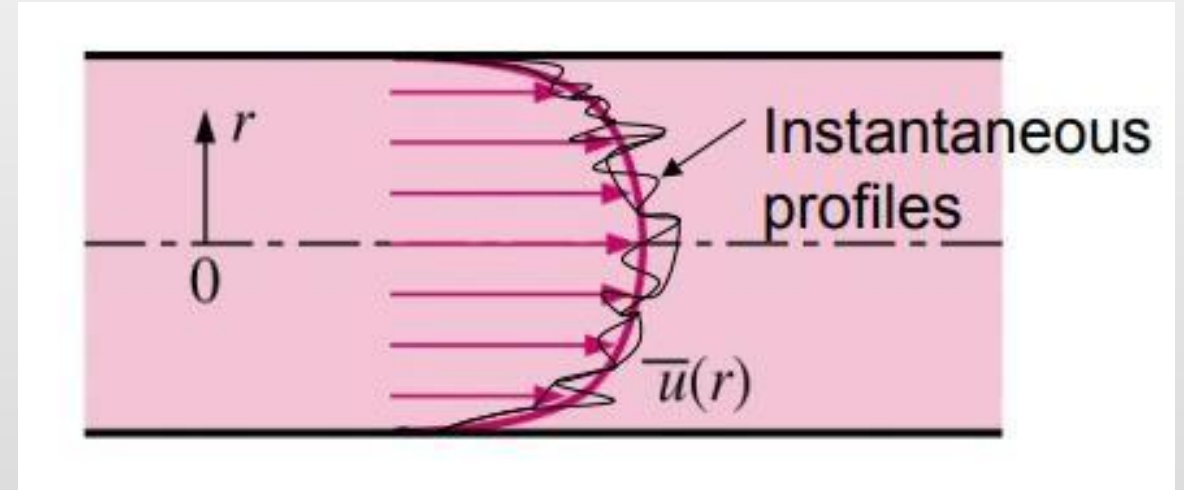
* Smooth surface. All values are for $Re = 10^6$ and are calculated from the Colebrook equation.

Turbulent Velocity Profile

Laminar velocity profile



Turbulent velocity profile



$U_{\text{mean}} = 0.5$ of U_{max} for Laminar Flow

&

$U_{\text{mean}} = 0.85$ of U_{max} for Turbulent Flow

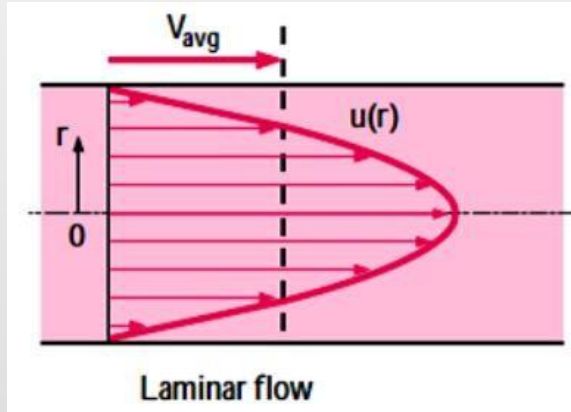
For example, if $U_{\text{mean}} = 1$,

(can vary with variation of Reynolds number)

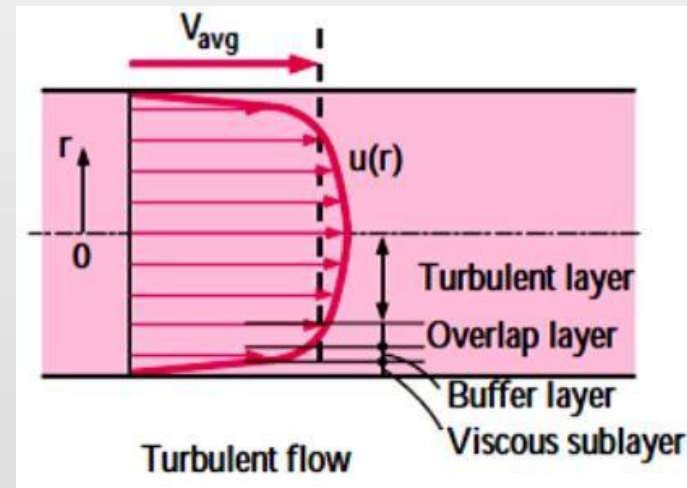
then $U_{\text{max}} = 2$ for Laminar flow and $U_{\text{max}} = 1.17$ for turbulent flow

Turbulent Velocity Profile

- Laminar flow has single layer, while turbulent flow has four layers of flow.



$$u(r) = 2V_{avg} \left(1 - \frac{r^2}{R^2} \right)$$



$$\frac{u}{u_{max}} = \left(1 - \frac{r}{R} \right)^{1/n}$$

$n = 7$ for most of the flows

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{dp}{dx} + \nu \frac{\partial^2 u}{\partial y^2}$$

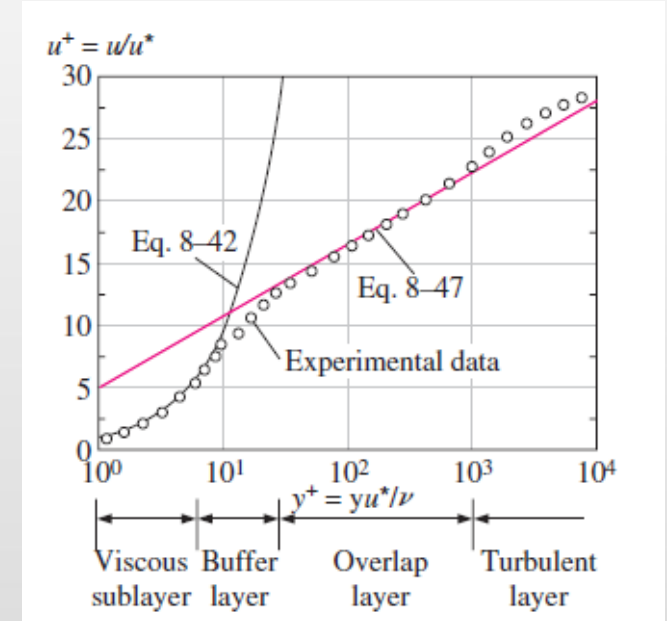
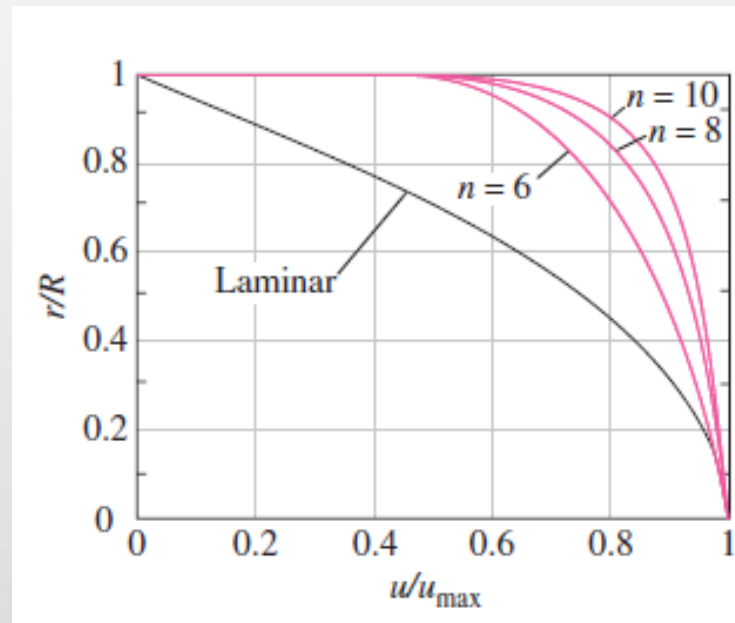
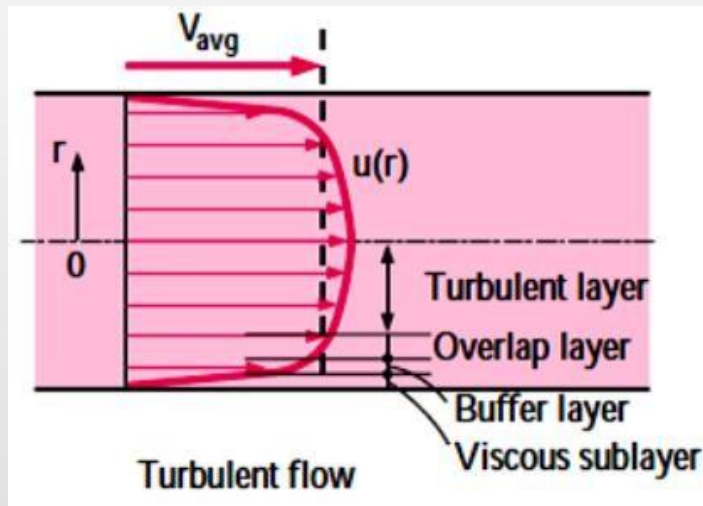
$$\tau_{turb} = -\rho \overline{u'v'} = \mu_t \frac{\partial \bar{u}}{\partial y}$$

$$\tau_{total} = (\mu + \mu_t) \frac{\partial \bar{u}}{\partial y} = \rho(\nu + \nu_t) \frac{\partial \bar{u}}{\partial y}$$

$$\frac{\partial}{\partial y} \left(\mu \frac{\partial \bar{u}}{\partial y} - \rho \overline{v'u'} \right) = \frac{\partial}{\partial y} \tau_{yx} = \frac{\partial}{\partial y} \left[\rho(\nu + \epsilon_m) \frac{\partial \bar{u}}{\partial y} \right]$$

where $\nu_t = \mu_t / \rho$ is the **kinematic eddy viscosity** or **kinematic turbulent viscosity** (also called the *eddy diffusivity of momentum*).

Turbulent Velocity Profile



$$\frac{u}{u_{\max}} = \left(1 - \frac{r}{R}\right)^{1/n}$$

$n = 7$ for most of the flows

Re =	4.0×10^3	2.3×10^4	1.1×10^5	1.1×10^6	2.0×10^6	3.2×10^6
$n =$	6.0	6.6	7.0	8.8	10	10

$$u^+ = y^+$$

Viscous
sublayer

$$\frac{u}{u_*} = 2.5 \ln \frac{yu_*}{\nu} + 5.0$$

Overlap or log layer

$$\frac{u_{\max} - u}{u_*} = 2.5 \ln \frac{R}{R - r}$$

Fully turbulent layer

$$y^+ = \frac{yu_*}{\nu} \quad \text{and} \quad u^+ = \frac{u}{u_*}$$

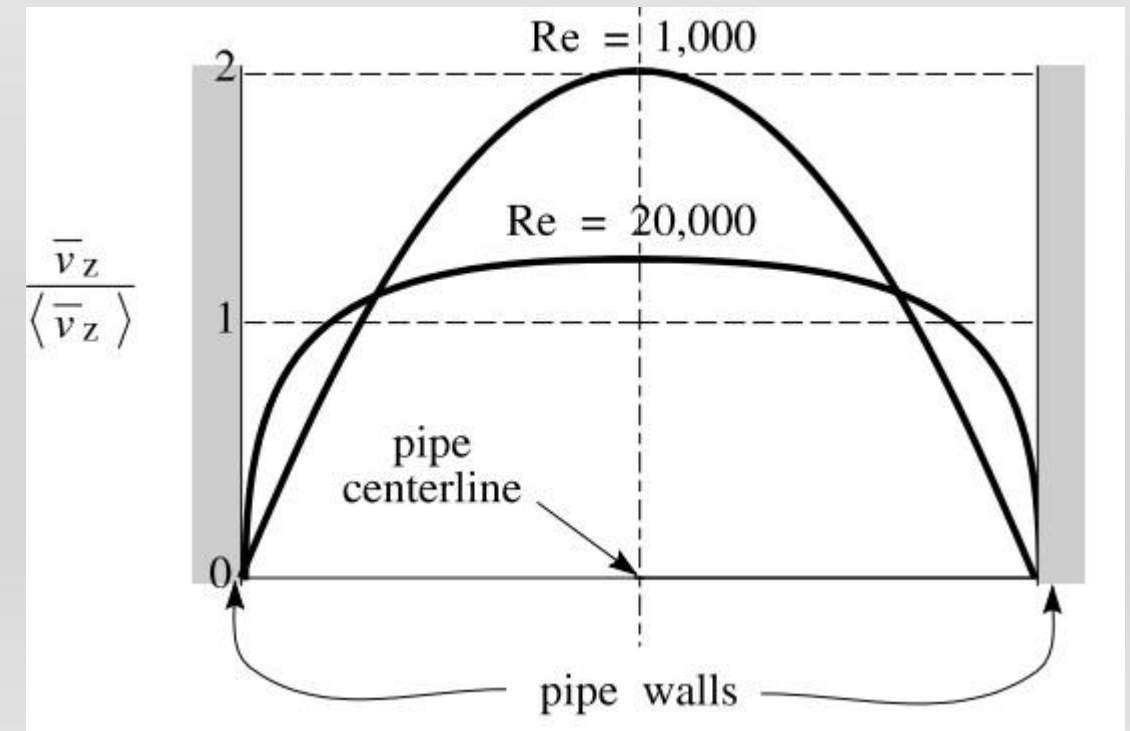
$y^+ = 1-5$ Sublayer
 $= 5-30$ buffer layer
 $= 30-1000 +$ for overlap and turbulent layer

Ratio of u_{mean} and u_{max} and value of n

- The reduction in the ratio of maximum to average velocity reflects the flattening of the profile as n becomes larger. Of course, this equation gives the kink in the profile at $r = 0$, and predicts the infinite slope at $r = R$, so it shouldn't be applied too close to either boundary, although it gives a reasonable fit otherwise.

$$\frac{u}{u_{\text{max}}} = \left(1 - \frac{r}{R}\right)^{1/n}$$

Re	n	$v_{\text{max}}/\langle v_z \rangle$
4.0×10^3	6	1.26
2.3×10^4	6.6	1.24
1.1×10^5	7	1.22
1.1×10^6	8.8	1.18
2.0×10^6	10	1.16
3.2×10^6	10	1.16



Analytical/Semi Empirical Velocity Profile

- Value of n , which is measure of curvature of profile is calculated to be 6.189

x	y
4000	6
23000	6.6
10000	6.189473684210

x	y
4000	1.26
23000	1.24
10000	1.253684210526

$$\frac{u}{u_{\max}} = \left(1 - \frac{r}{R}\right)^{1/n}$$

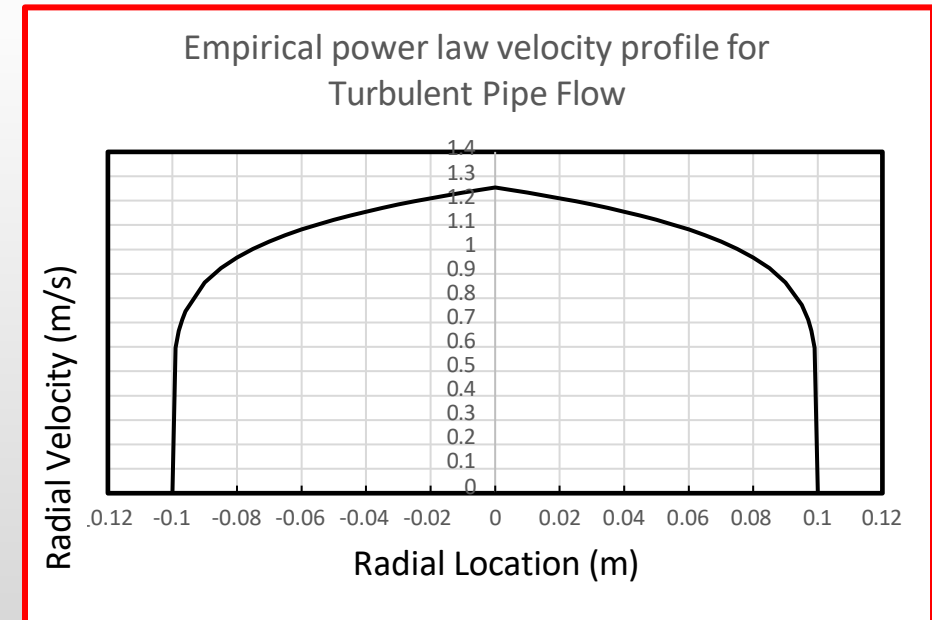
- Ratio of $u_{\max}$ to u_{mean} for current case is : 1.254 or $u_{\max} = 1.254 u_{\text{mean}}$

- Therefore, equation for empirical profile becomes:

$$u(r) = 1.254 u_{\text{mean}} \left(1 - \frac{r}{R}\right)^{1/6.189}$$

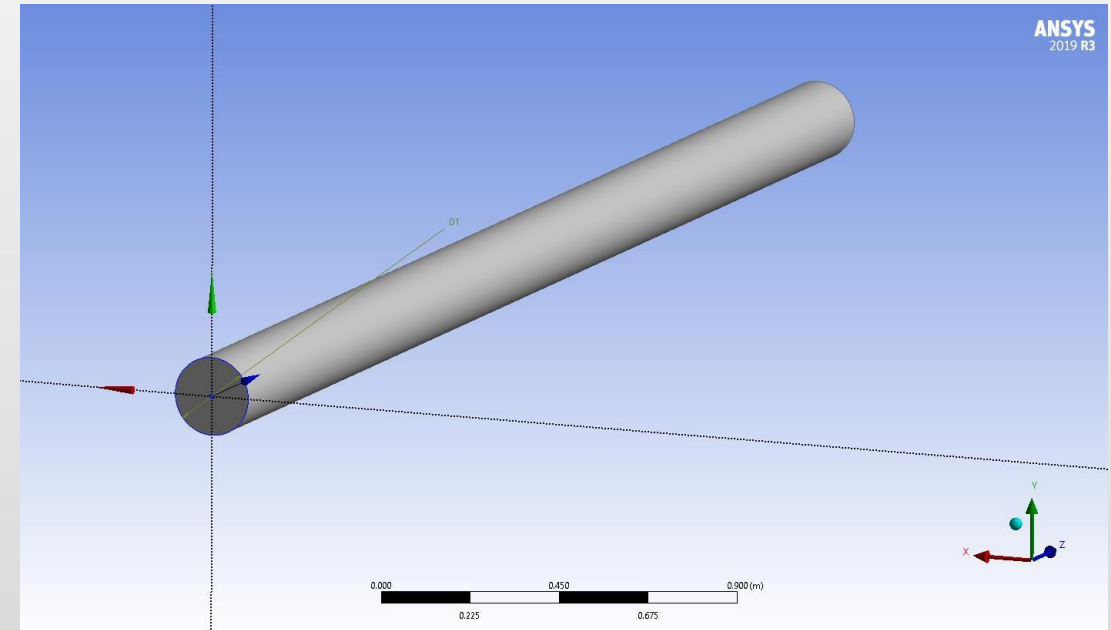
Where r varies from 0 to R ($R = \frac{0.2}{2} = 0.1$)

r	u(r)
-0.1	0
-0.099	0.5358531
-0.098	0.6664738
-0.097	0.7115391
-0.096	0.745457
-0.095	0.7728248
-0.09	0.8644115
-0.085	0.9229387
-0.08	0.9668521
-0.075	1.0023479
-0.07	1.0323153
-0.065	1.0583503
-0.06	1.0814329
-0.055	1.1022108
-0.05	1.1211353
-0.045	1.1385343
-0.04	1.154654
-0.035	1.1696842
-0.03	1.1837744
-0.025	1.1970445
-0.02	1.2095926
-0.015	1.2214934
-0.01	1.2328328
-0.005	1.24365
0	1.254
0.005	1.24365
0.01	1.2328328
0.015	1.2214934
0.02	1.2095926
0.025	1.1970445
0.03	1.1837744
0.035	1.1696842
0.04	1.154654
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0.07	1.0323153
0.075	1.0023479
0.08	0.9668521
0.085	0.9229387
0.09	0.8644115
0.095	0.7728248
0.096	0.745457
0.097	0.7115391
0.098	0.6664738
0.099	0.5358531
0.1	0



Geometry

- Design modeler
- Length = 16 meter and $D = 0.2$, $L/D = 80$
- Full model is modeled.
- You can try with 90 deg or 30 deg sector.



Meshing

- Meshing in ANSYS Meshing
- $V = 1 \text{ m/s}$, $\rho = 1000 \text{ kg/m}^3$, $\mu = 0.02 \text{ kg/(m s)}$
- $L = 16 \text{ m}$, $D = 0.2 \text{ m}$
- $Re = 10,000$
- For first cell height, I will use inbuilt Y+ calculator
- $C_f = 0.079 / Re^{(1/5)}$;

$$C_f = 0.079 / (10000)^{(0.25)} = 0.079 / 10 = 0.0079$$

- $c_f = \tau_w / (0.5 * \rho * V_{avg}^2)$
- $\tau_w = 0.5 * \rho * u^2 * c_f = 0.5 * 1000 * 0.0079 = 3.95 \text{ Pa}$
- $c_f = f / 4$
- Friction velocity $u\tau = \sqrt{(\tau_w / \rho)} = 0.062849 \text{ m/sec}$

Meshing

$$V = 1 \text{ m/s}, \rho = 1000 \text{ kg/m}^3, \mu = 0.02 \text{ kg/(m s)}$$

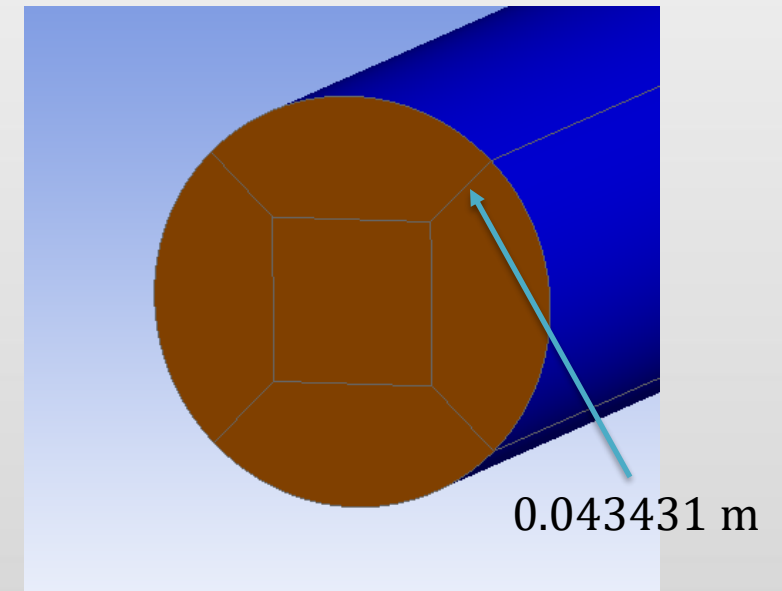
$$L = 16 \text{ m}, D = 0.2 \text{ m}$$

- $y^+ = (y * u\tau * \rho) / \mu$
- $y = (y^+ * \mu) / (u\tau * \rho) = (1 * 0.02) / (0.062849 * 1000) = 0.000318229 \text{ m} = 0.318 \text{ mm}$
- First cell height = 0.000318229 meters or 0.318 mm

Meshing

In ansys meshing (sweep method) , I cannot directly set the first cell height. Instead, I have option to provide the bias factor, number of division and length of edge

- $L = \sum l_1 * r^i = l_1 * (1 - r^n) / (1 - r)$
- $L = \text{Edge length} = 0.043431 \text{ m}$
- $n = \text{number of divisions} = 22$
- $l_1 = \text{First cell height}$



Meshing

Meshing Formulas

The following equations govern the distribution of elements along the edge to ensure accurate boundary layer simulation:

- **Bias Factor (BF):**
$$BF = l_n / l_1 = r^{n-1}$$
- **Growth Rate (r):**
$$r = (BF)^{1/(n-1)}$$
- **First Element Length (l_1):**
$$l_1 = [L * (1 - r)] / (1 - r^n)$$

3. Terminology

Term	Definition
l_1	First element length
l_n	Last element length
r	Growth rate
n	Number of divisions
L	Total edge length

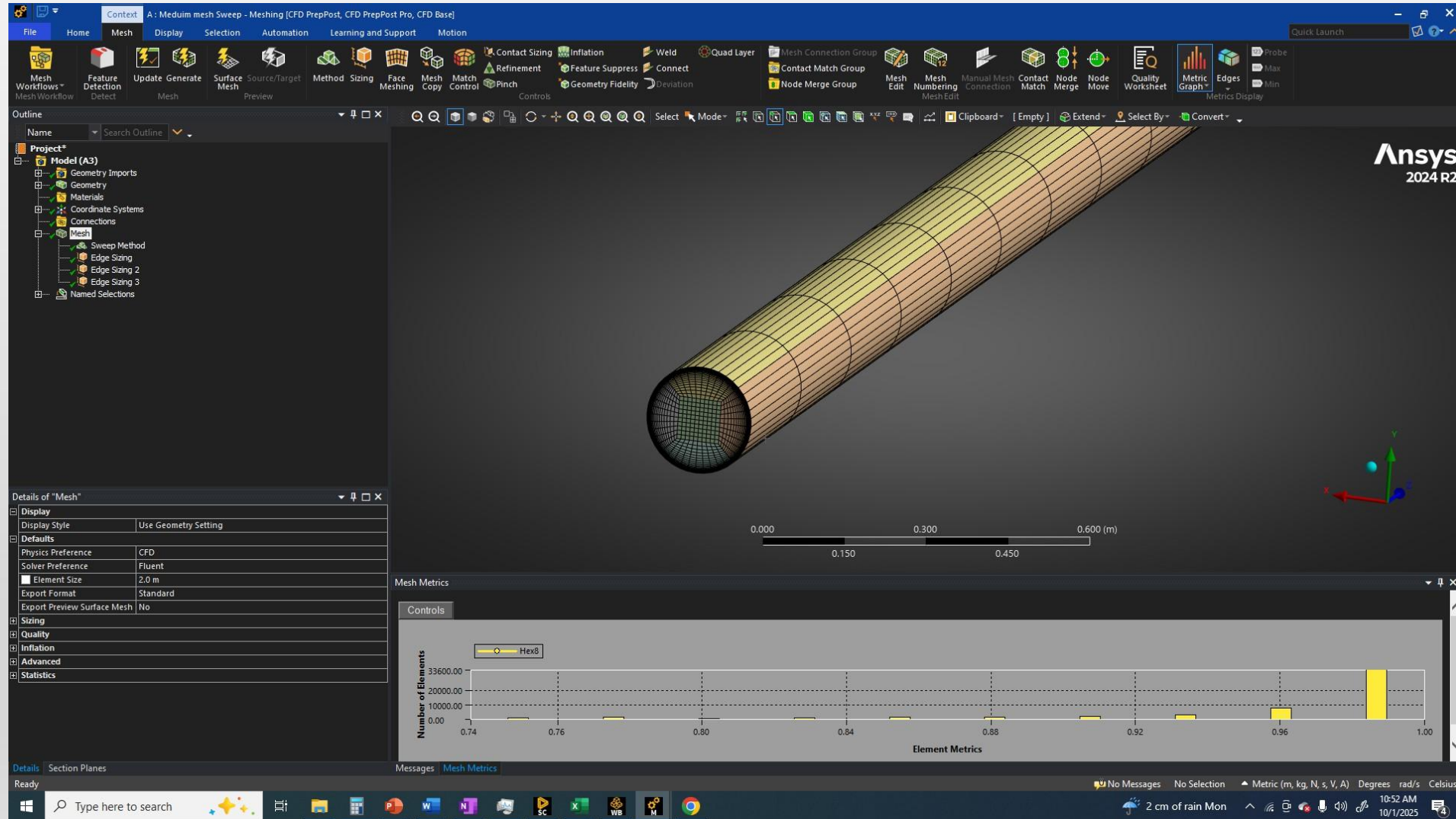
Meshing

Calculated Parameters for Turbulent Pipe Flow

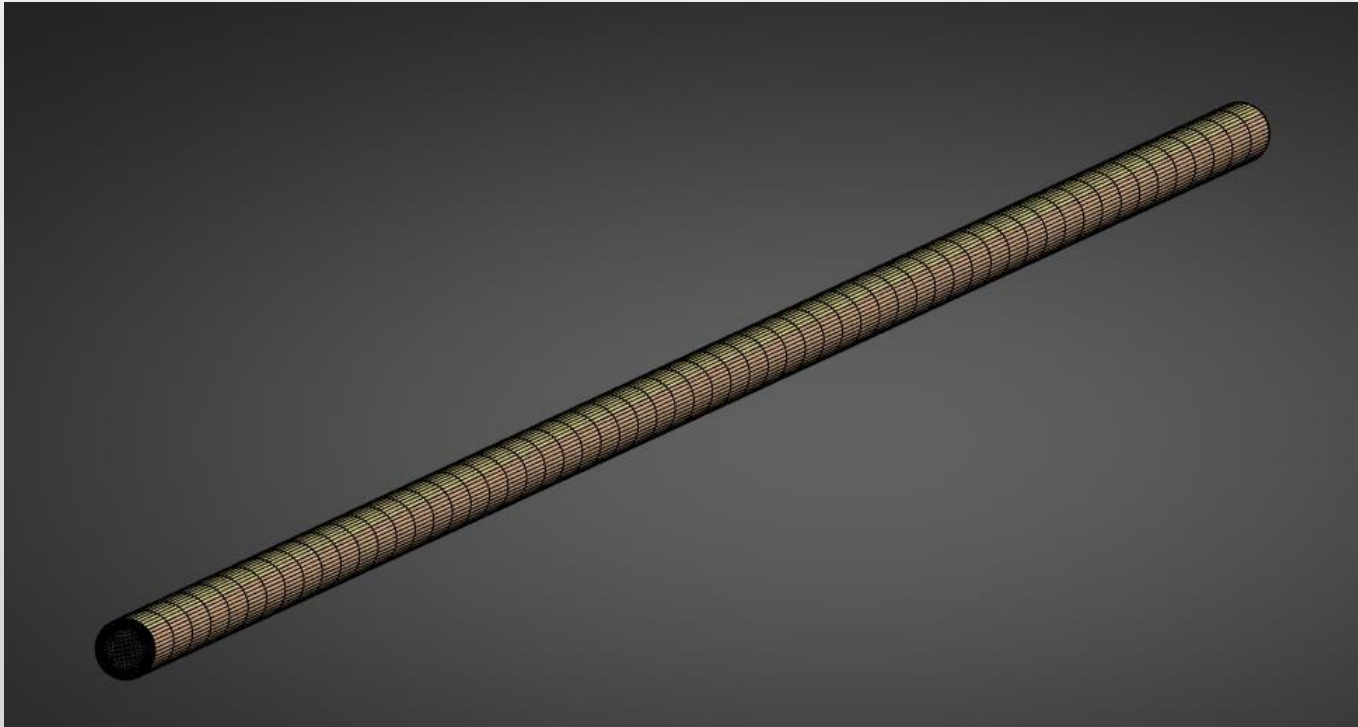
Based on the CFD validation requirements ($Re = 10,000$), the following parameters were utilized for the mesh generation:

- **Target First Cell Height (l_1):** 0.000318 m
- **Edge Length (L):** 0.043431 m
- **Growth Rate (r):** 1.15
- **Number of Divisions (n):** 22
- **Bias Factor (BF):** 19

Meshing



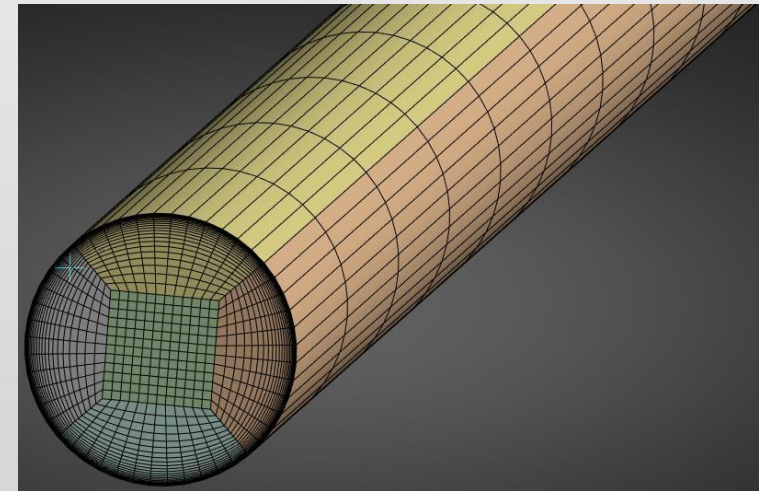
Meshing



Number of cells = 72000

Min orthogonal quality = 0.735

First cell height = 0.000318 m for $Y^+ \sim 1$
= 0.00318 m for $Y^+ \sim 10$



Results

- Darcy friction factor f at 14 m

	f	Error %
CFD	0.0325	5.22 %
Empirical (S.E. Haaland 1983)	0.0308	

Results

- Coefficient of friction C_f at 14 m

	C_f	Error %
CFD	0.00812	5.18 %
Empirical (S.E. Haaland 1983)	0.00772	

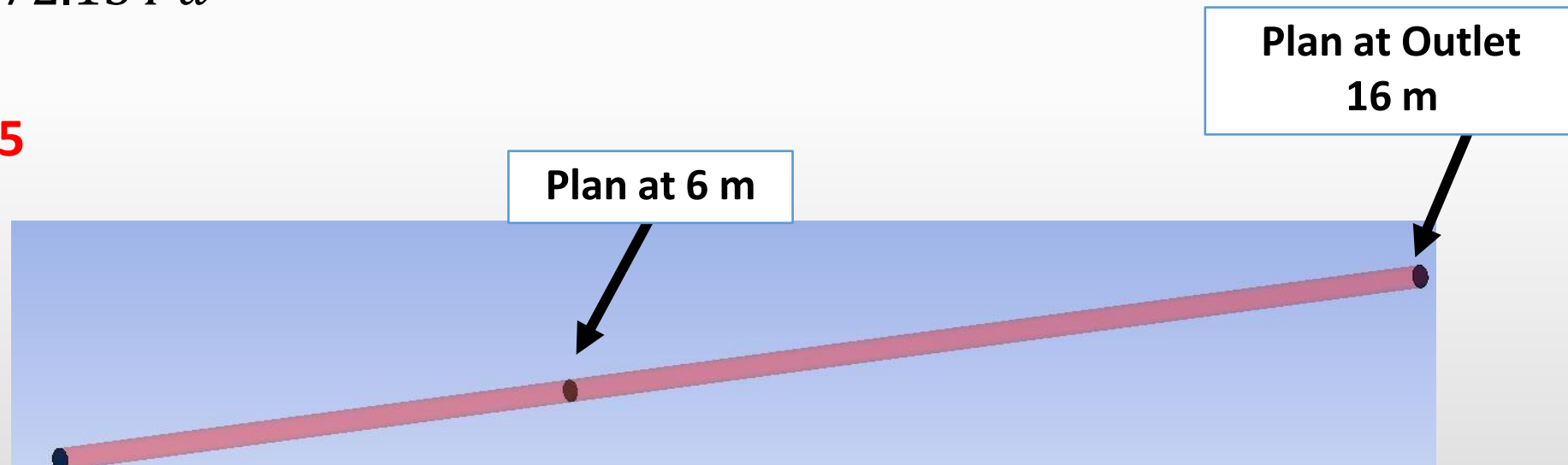
Results

- **Pressure drop (from 6 m to 16 m)**

$$\Delta P_{CFD} = 811.705 \text{ Pa}$$

$$\Delta P_{Empirical} = 772.15 \text{ Pa}$$

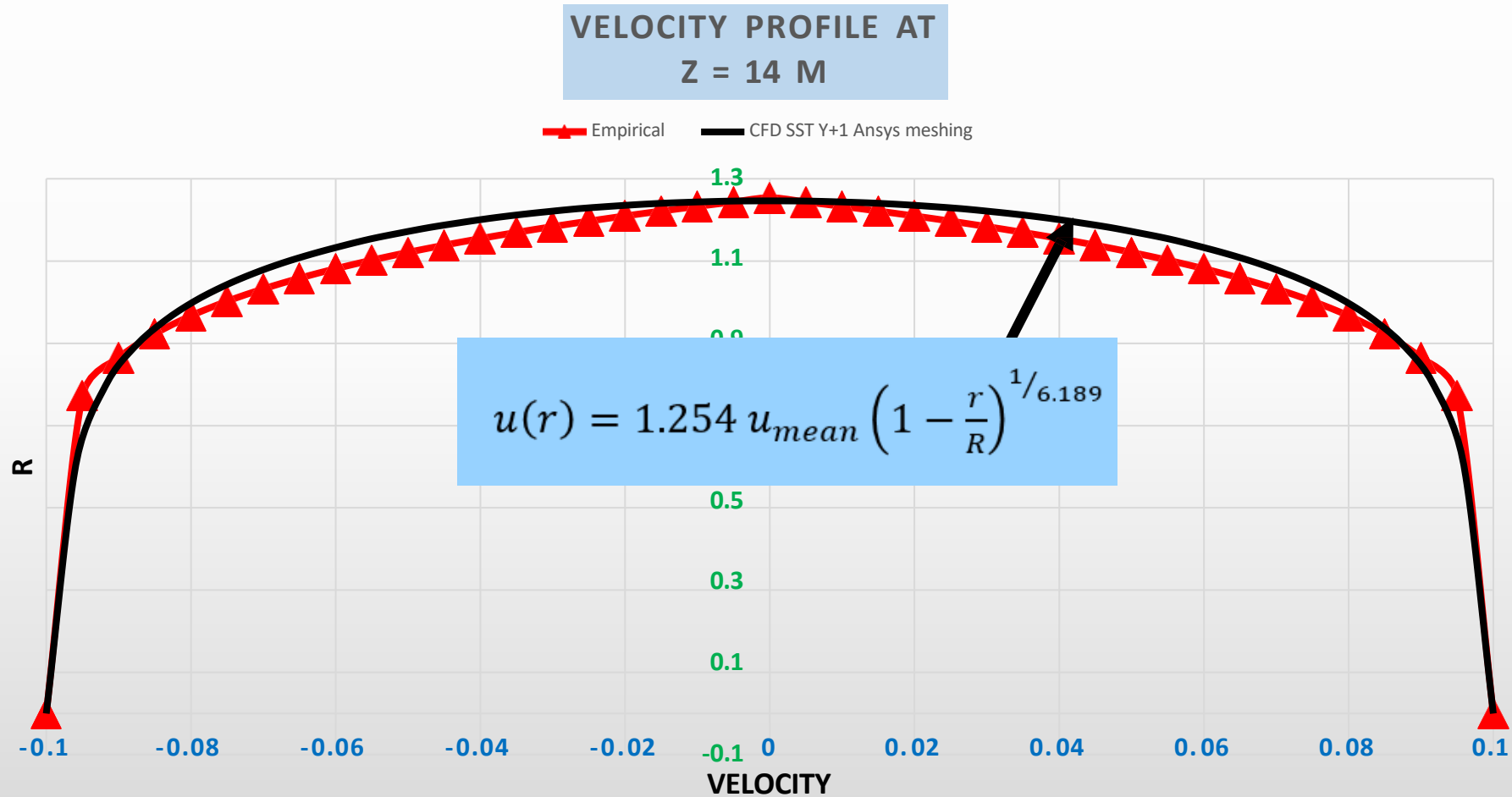
Error % = 5



Please note, I have compared the pressure drop in fully developed region as empirical formulas are valid only in that region.

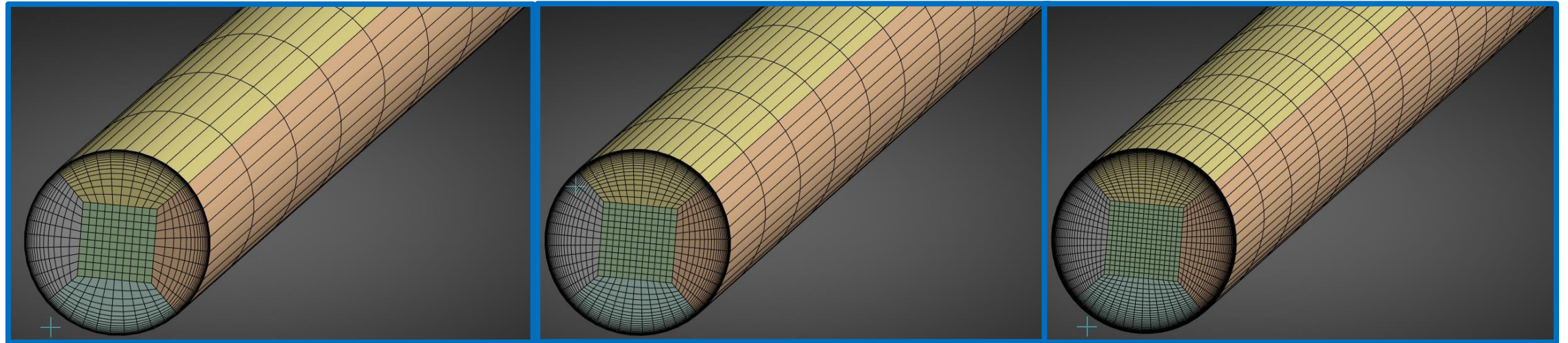
Velocity profile with Medium Mesh

$Y+ = 1$, SST model



Mesh independence Study

- Three meshes for mesh independence study. I have used different BF for three meshes to keep the first cell height = 0.000318 m (**For $Y^+ < 1$**)



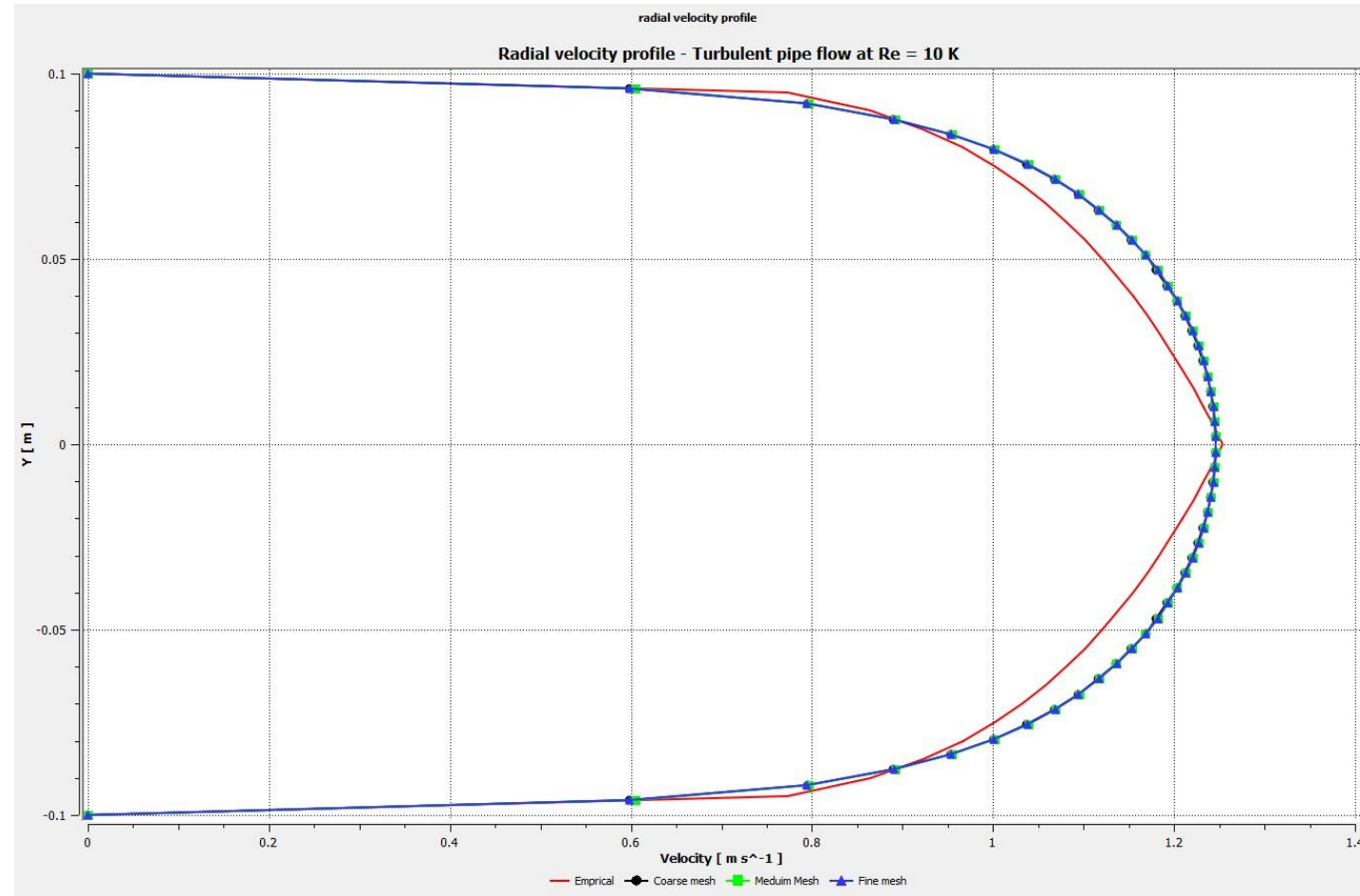
Coarse
37440 Cells

Medium
72000 Cells

Fine
142875 Cells

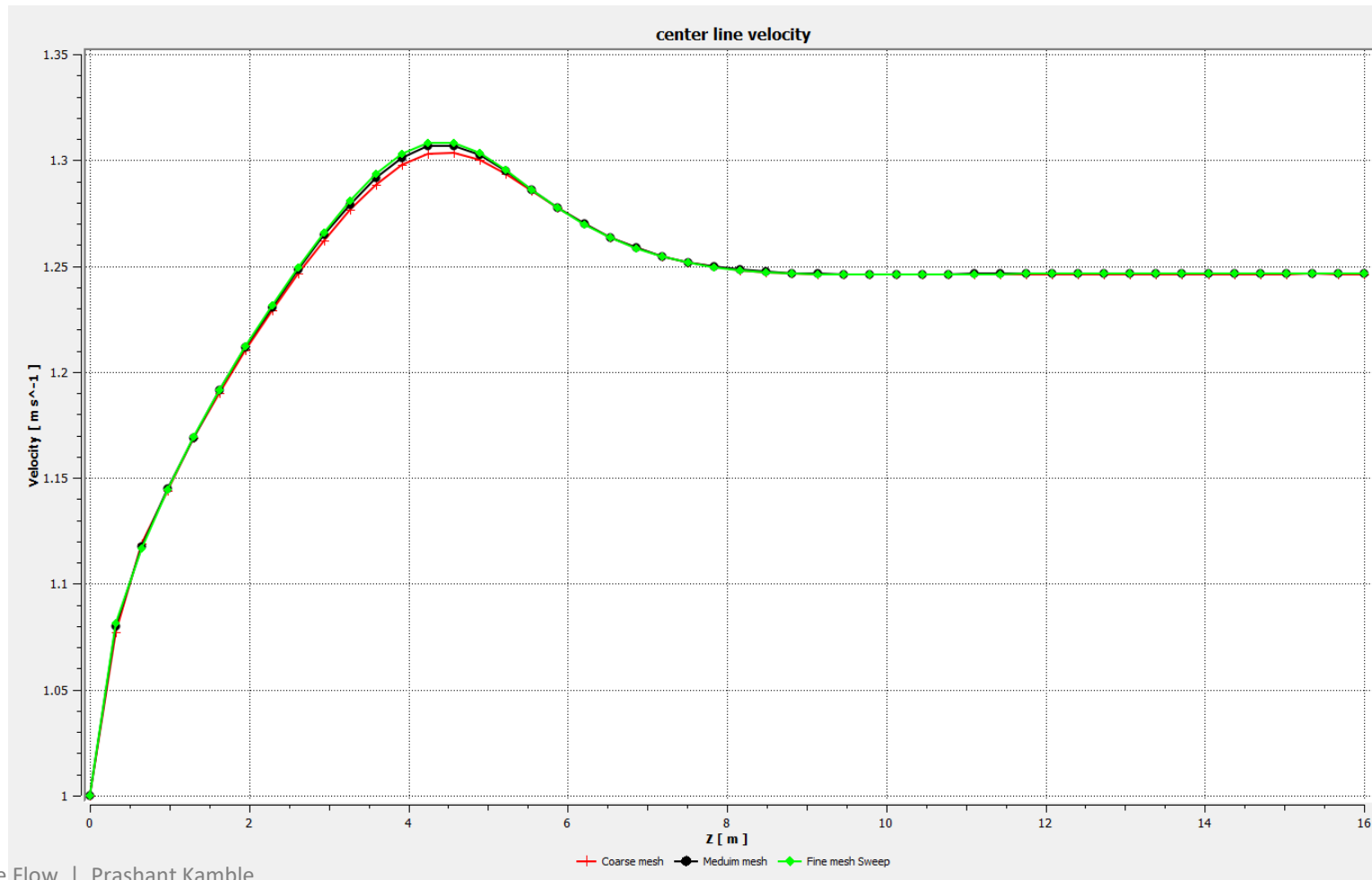
Velocity profile at $Z = 14$ m for three meshes

- SST Turbulence model with $Y^+ \sim 1$

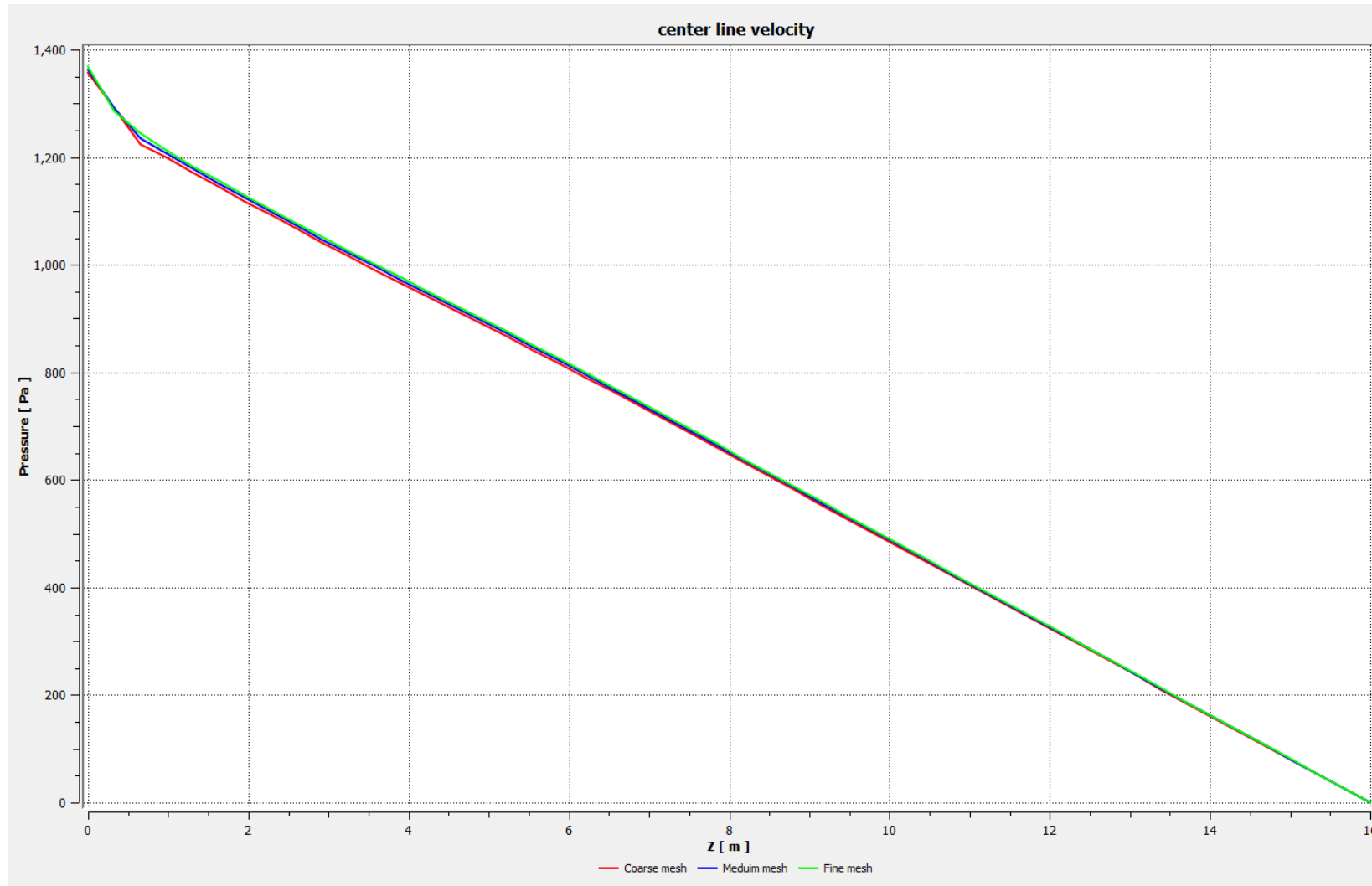


Axial velocity profile for three meshes

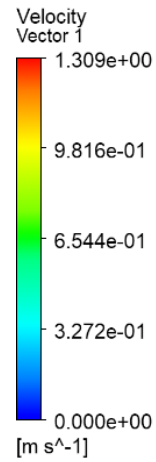
- SST Turbulence model with $Y^+ \sim 1$



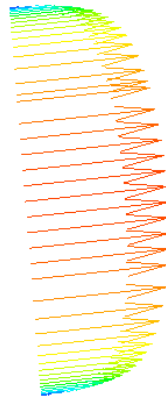
Pressure drop for three meshes



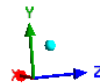
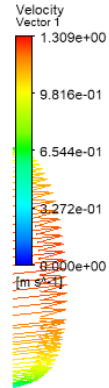
Velocity Vectors



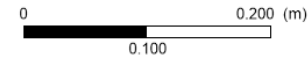
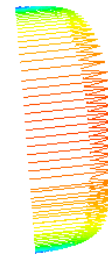
Coarse



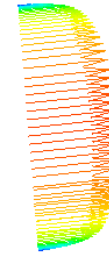
Ansys
2024 R2



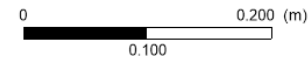
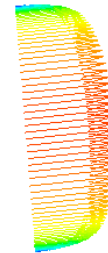
Medium



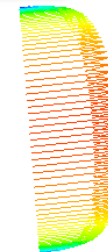
Ansys
2024 R2



Fine



Ansys
2024 R2



Turbulence model study

- SST vs RKE for $Y^+ \sim 1$, Medium mesh

	CFD (RKE)	CFD (SST)	Empirical
Darcy friction factor f	0.0341	0.0325	0.030886
Coefficient of friction C_f	0.00853	0.00812	0.00772
Pressure drop ΔP	855.16	811.705	772.15

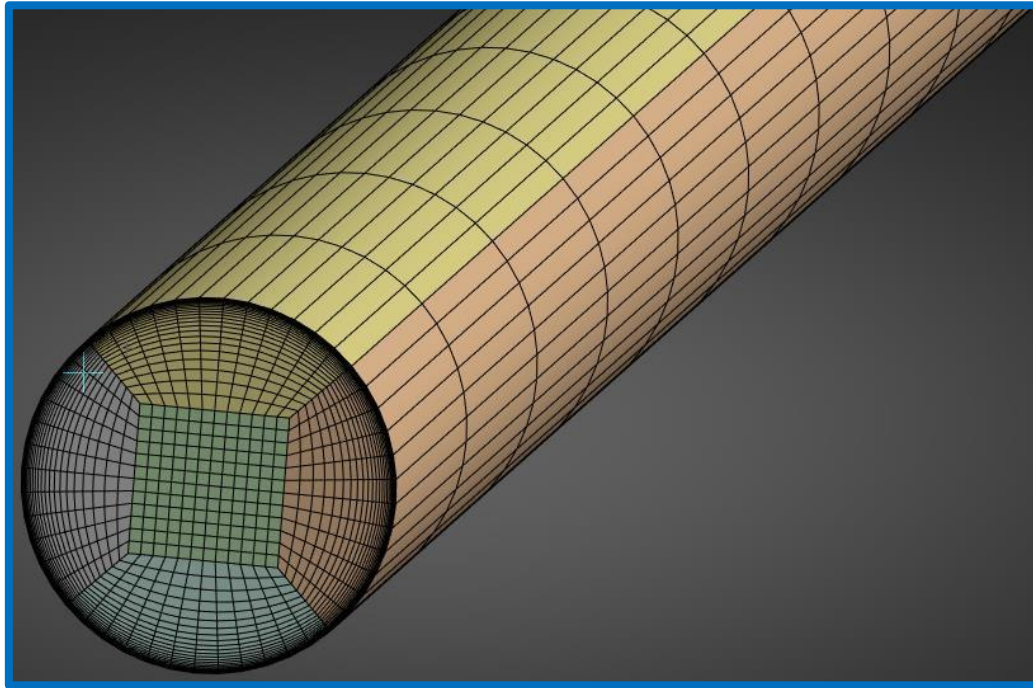
Y+ effect on results

- SST for Y+ ~ 1 & 10, Medium mesh

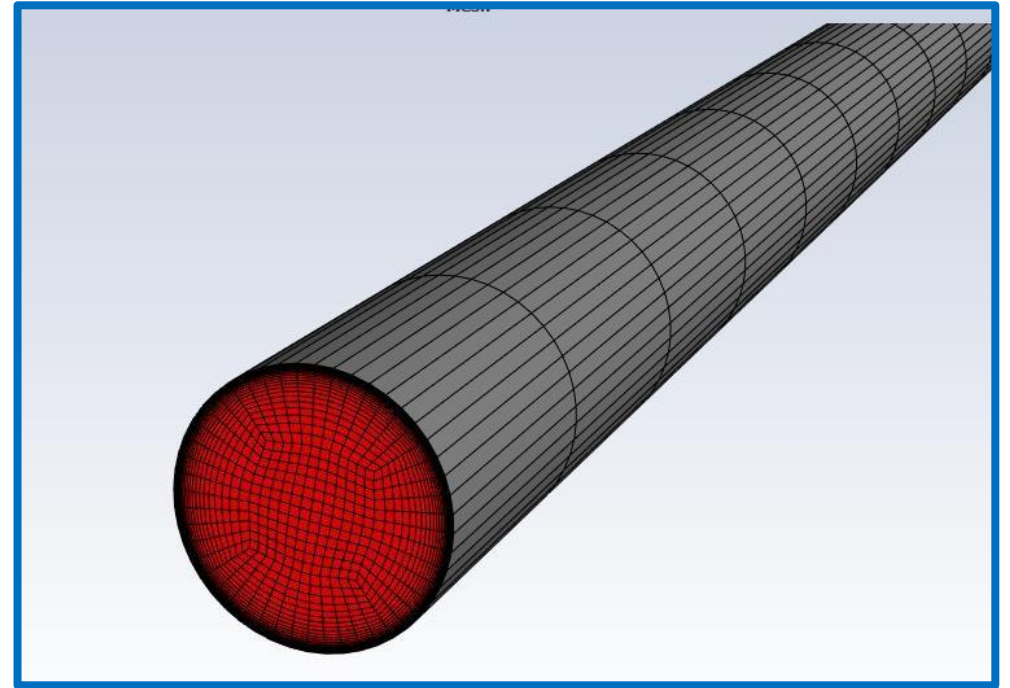
	CFD (Y+ 1)	CFD (Y+10)	Empirical
Darcy friction factor f	0.0325	0.0327	0.030886
Coefficient of friction Cf	0.00812	0.00818	0.00772
Pressure drop ΔP	811.705	811.705	772.15

ANSYS Workbench Mesh and ICEMCFD Mesh

- Both have same number of cells i.e. 72000

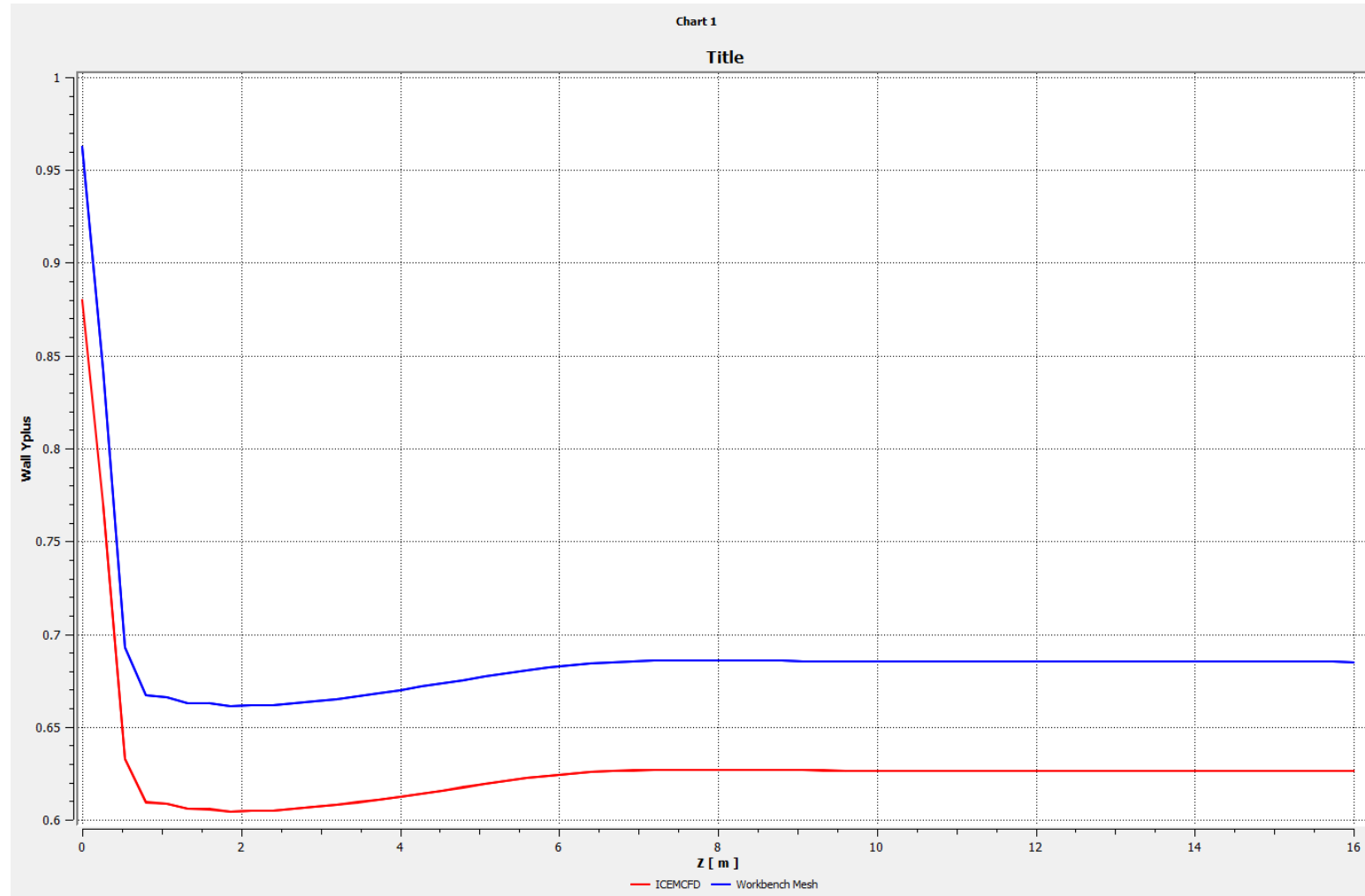


ANSYS Workbench
Min ortho quality ~ 0.779



ICEMCFD
Min ortho quality ~ 0.822

Y plus graph for both meshes

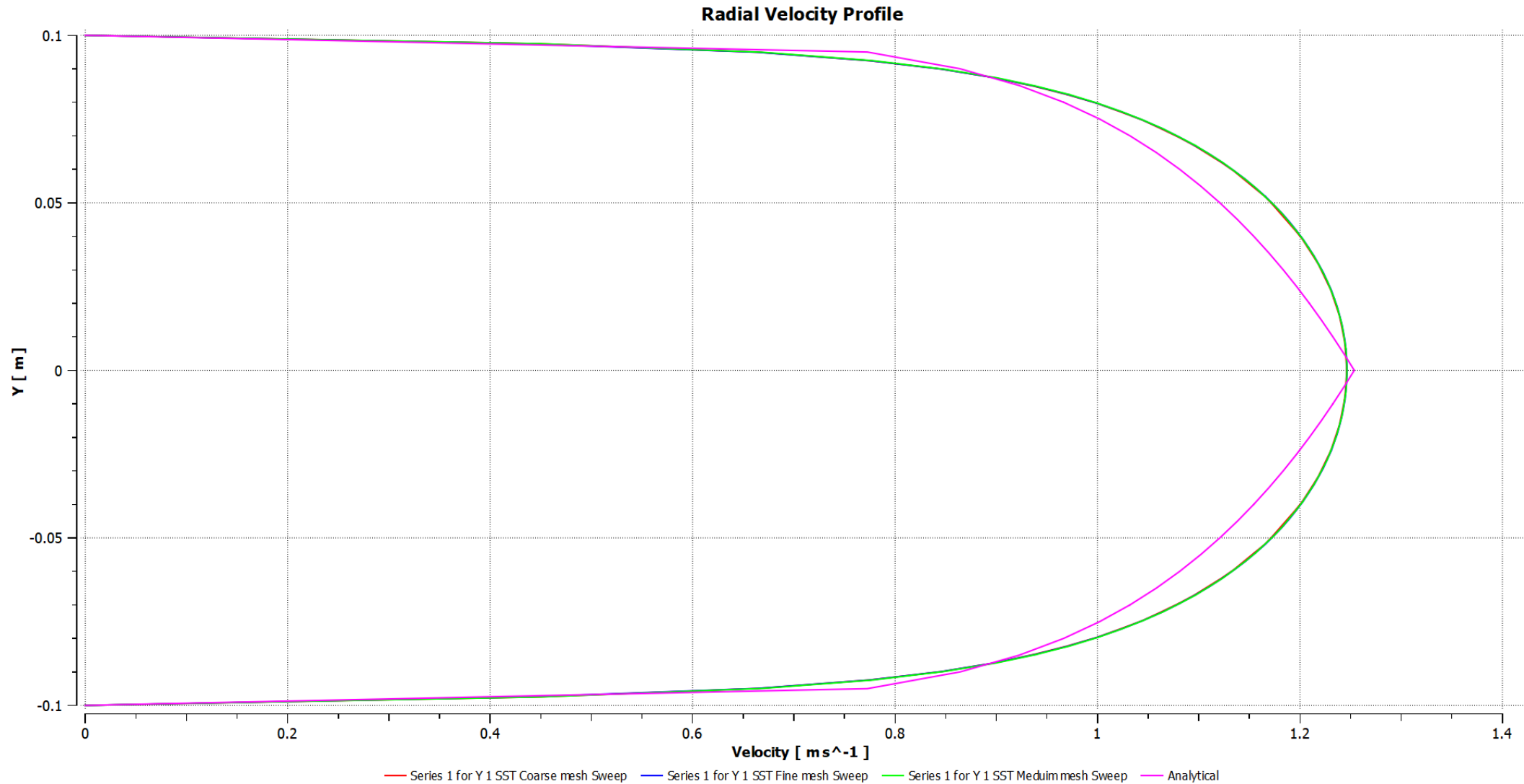


Effect of different meshing Codes

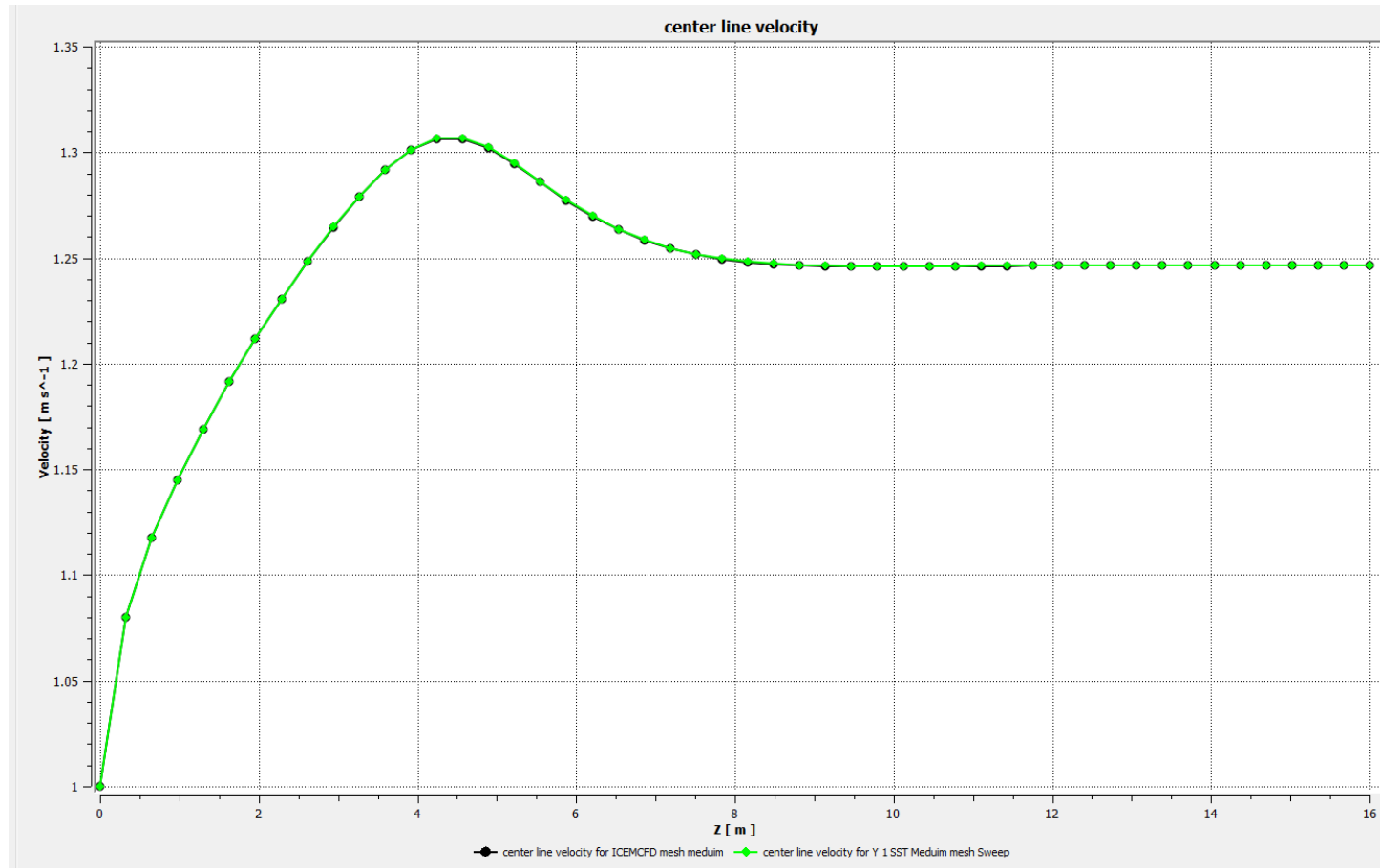
- SST model, Y+1 , Medium mesh

	CFD (ANSYS Workbench)	CFD (ICEMCFD)	Empirical
Darcy friction factor f	0.0325	0.0325	0.030886
Coefficient of friction C_f	0.00812	0.00812	0.00772
Pressure drop ΔP	811.705	812.002	772.15

Radial velocity profile comparison



Center line velocity

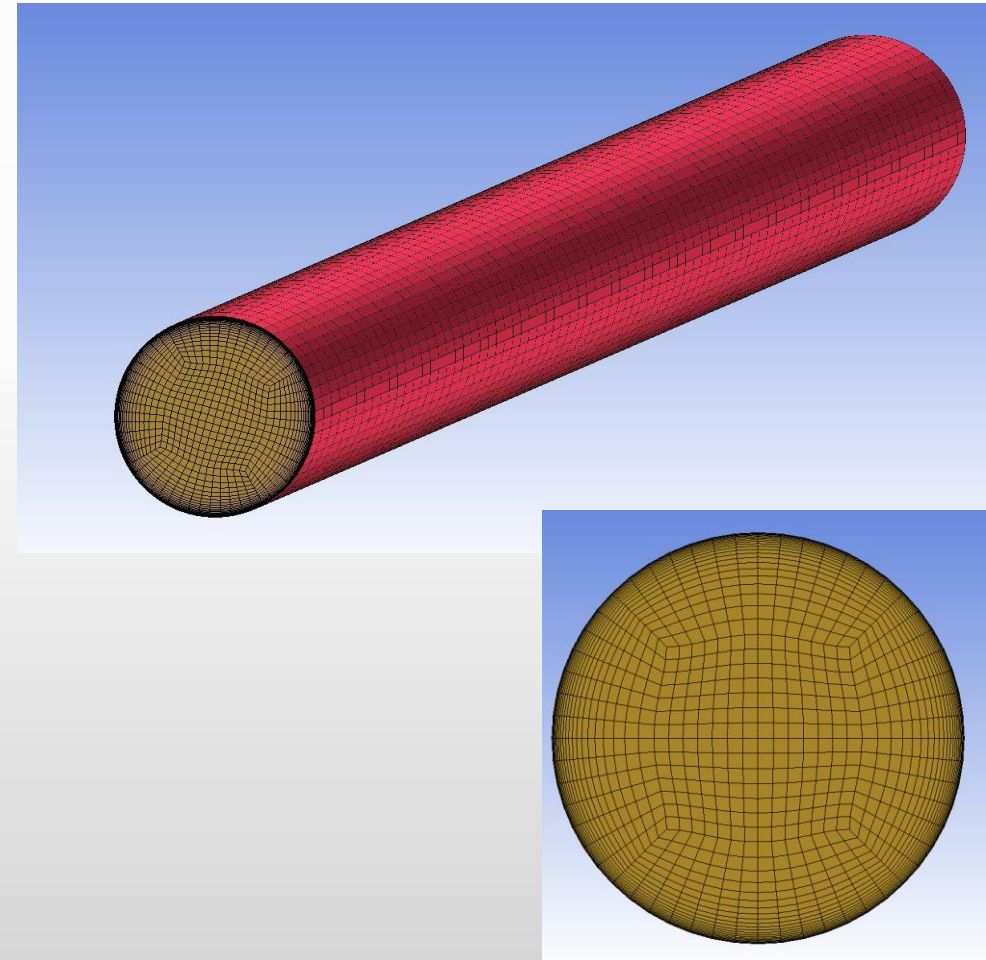


CFD with ICEMCFD Hexa Meshing

Hexa Mesh in ICEMCFD

- First cell height = 0.00029 m (For $Y^+ \sim 1$)
- Total number of cells = 115668 cells

First Cell Hieght Calculator (Cf Correlation 1)				
$\Delta y_1 = \frac{Y^+ \cdot \mu}{u_\tau \cdot \rho}$		$R_e = \frac{\rho V D}{\mu}$		
Input				
Inputs/Sr. No.	Parameters	Value	Full Units	units
1	Diameter	0.2	meters	m
2	Reynolds Number R_e	10000	-	-
3	Friction Velocity u_τ	0.067795	Meters per second	m/s
4	Non dimensional Wall Distance Y^+	1	-	-
5	Density ρ	1000	Kg/m3	kg/m3
6	Viscosity μ	0.02	Pascal-Seconds	Pa.s
7	Velocity u	1	Meters per second	m/s
Output :		Δy_1		
ΔY_1	=	0.0002950	Meters	



Comparison of Coefficient of Friction

S.E. Haaland 1983

- Friction factor for smooth pipe for current case

$$f = 0.030886$$

Please note that from Colebrook equation, same results were obtained

$$\frac{1}{\sqrt{f}} \cong -1.8 \log \left[\frac{6.9}{\text{Re}} + \left(\frac{\varepsilon/D}{3.7} \right)^{1.117} \right]$$

$$C_f = \frac{f}{4}$$

- So $C_f = 0.007215$ (Empirical)
- From CFD $C_f = 0.00857$ (For Realizable K-Epsilon model for $Y^+ \sim 1$ ICEMCFD mesh)
- $\% \text{ Error} = \frac{0.00857 - 0.007215}{0.007215} \times 100 = 18\%$

Conclusion:

SST k- ω Model Validation: Highly accurate for turbulent pipe flow at $Re = 10,000$, validating within 5% of Haaland's empirical correlation.

Turbulence Model Comparison: Realizable k- ϵ (RKE) model yields a higher deviation ($\sim 11\%$ error) compared to SST under identical mesh configurations.

Near-Wall Resolution: Resolving the viscous sublayer ($Y^+ \sim 1$) is critical for accurate near-wall velocity and friction coefficient predictions.

Mesh Independence: Achieved at a medium mesh resolution (72,000 cells), showing no significant change in bulk parameters compared to the fine mesh (142,875 cells).

Meshing Code Consistency: Mesh quality and code comparison show consistent pressure drop predictions between ANSYS Workbench and ICEMCFD.

Conclusion

SST k- ω Model Validation: Displays excellent predictive accuracy for turbulent pipe flow at $Re = 10,000$, validating within a 5% error margin against Haaland's empirical correlation.

Turbulence Model Comparison: The Realizable k- ϵ (RKE) model yields a larger deviation ($\sim 11\%$ error) compared to the SST k- ω model under identical mesh parameters.

Near-Wall Grid Resolution: Resolving the viscous sublayer ($y^+ \approx 1$) is critical for capturing steep velocity gradients and predicting skin friction coefficients accurately.

Grid Convergence: Mesh independence is established at the medium mesh level (72,000 cells), with no significant change in bulk parameters compared to the fine mesh (142,875 cells).

Solver & Mesh Portability: Excellent agreement is observed between ANSYS Workbench and ICEMCFD meshes, demonstrating numerical robustness across different meshing platforms.